1. $a_p = \frac{d_w - d_m}{2} = 4mm$　$V_f = f \cdot n$ => $f = \frac{V_f}{n} = 0.4mm/r$

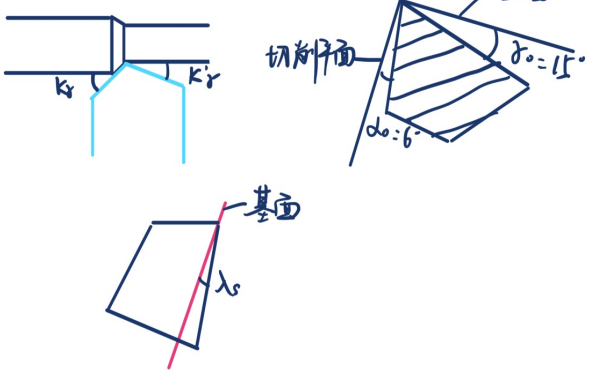
$V_c = \frac{\pi \cdot d_w \cdot n}{1000} = 46.75 m/min$　$Q = 1000 V_c f a_p = 74800 mm^3/min$

2. ①高速钢具有极好的淬透性，较高的强度和优异的韧性，最关键的是其可加工性极好，适合制造形状复杂的刀具

③YT类硬质合金导热性较差，且不锈钢及高温合金材料中常含钛族，切削时温度高且容易与工件发生亲和导致严重粘刀，加剧刀具磨损，不宜选用。

③ 粗铣铸铁箱体平面　YG类
精镗铸铁箱体孔　YG类
齿轮加工向剃齿刀　高速钢
45钢棒料粗加工　YT类
精车40Cr工件外圆　YT类

3. 主偏角、副偏角　基面
刃倾角　切削平面
前角、后角　主剖面



$\lambda_s = 15°$ 时　$\cos\lambda_s = 0.965$

$\tan\gamma_n = \tan\gamma_o \cos\lambda_s = 0.388$　$\gamma_n = 14.5°$

$\cot\alpha_n = \cot\alpha_o \cos\lambda_s = 9.19$　$\alpha_n = 6.2°$

$\tan\gamma_p = \tan\gamma_o \cos k_r + \tan\lambda_s \sin k_r$　$\gamma_p = 20.1°$

$\cot\alpha_p = \cot\alpha_o \cos k_r + \tan\lambda_s \sin k_r$　$\alpha_p = 11.3°$

$\lambda_s = 15°$ 时　$\cos\lambda_s = 0.965$

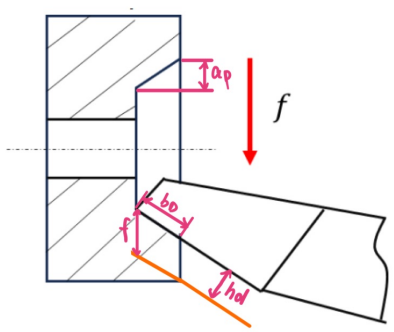
$\tan\gamma_n = \tan\gamma_o \cos\lambda_s = 0.388$　$\gamma_n = 14.5°$

$\cot\alpha_n = \cot\alpha_o \cos\lambda_s = 9.19$　$\alpha_n = 6.2°$

$\tan\gamma_p = \tan\gamma_o \cos k_r + \tan\lambda_s \sin k_r$　$\gamma_p = -5.6°$

$\cot\alpha_p = \cot\alpha_o \cos k_r + \tan\lambda_s \sin k_r$　$\alpha_p = 12.5°$

4.



$h_D = f \cdot \sin k_r = 0.29mm$

$b_D = \frac{a_p}{\sin k_r} = 3.11mm$